

COMBINATORIAL CATEGORICAL EQUIVALENCES; SIMPLIFIED

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1. INTRODUCTION

The intention of this paper is to prove a class of equivalences of categories that seem of interest in enumerative combinatorics as per [20], representation theory as per [10], and homotopy theory as per [3]. More specifically, for a class of categories $\mathcal{P}$, we construct a category $\mathcal{D}$ with zero morphisms (that is, $\mathcal{D}$ has homs enriched in the category $1/\text{Set}$ of pointed sets) and an equivalence of categories of the form

$$[\mathcal{P}, \mathcal{K}] \simeq [\mathcal{D}, \mathcal{K}]_{\text{pt}} . \quad (1.1)$$

On the left-hand side we have the usual category of functors from $\mathcal{P}$ into any additive category $\mathcal{K}$ which has finite direct sums and splitting for idempotents. On the right-hand side we have the category of functors which preserve the zero morphisms.

One example has $\mathcal{P} = \Delta_{\perp, \top}$, the category whose objects are finite ordinals with first and last element, and whose morphisms are functions preserving order and first and last elements. Then $\mathcal{D}$ is the category with non-zero and non-identity morphisms

$$0 \xleftarrow{\partial} 1 \xleftarrow{\partial} 2 \xleftarrow{\partial} \dots$$

such that $\partial \circ \partial = 0$. Since there is an isomorphism of categories

$$\Delta_{\perp, \top} \cong \Delta_+^{\text{op}} , \quad (1.2)$$

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where the right-hand side is the algebraist's simplicial category (finite ordinals and all order-preserving functions), our result (1.1) reproduces the Dold-Puppe-Kan Theorem [13, 14, 23].

Cubical sets also provide an example of our setting; see Example 3.3. We conclude that cubical simplicial abelian groups are equivalent to semi-simplicial abelian groups.

For a category $\mathcal{A}$ equipped with a suitable factorization system $(\mathcal{E}, \mathcal{M})$ [15], write $\text{Par}\mathcal{A}$ (strictly the notation should also show the dependence on $\mathcal{M}$) for the category with the same objects as $\mathcal{A}$ and with $\mathcal{M}$ -partial maps as morphisms. We identify $\mathcal{E}$ with the subcategory of $\mathcal{A}$ having the same objects but only the morphisms in $\mathcal{E}$. Assume each object of $\mathcal{A}$ has only finitely many $\mathcal{M}$ -subobjects. Let $\mathcal{X}$ be any additive category with finite direct sums and splitting idempotents. Our main equivalence (1.1) includes as a special case an equivalence of categories

$$[\text{Par}\mathcal{A}, \mathcal{X}] \simeq [\mathcal{E}, \mathcal{X}] . \quad (1.3)$$

(Here $\mathcal{D}$ is obtained from $\mathcal{E}$ by freely adjoining zero morphisms.) Indeed, we show that this special case implies the general case.

Let $\mathfrak{S}$ be the groupoid of finite sets and bijective functions. Let $\text{FI}\sharp$ denote the category of finite sets and injective partial functions. Let Mod^R denote the category of left modules over the ring R . Our original motivation was to understand and generalize the classification theorem for $\text{FI}\sharp$ -modules appearing as Theorem 2.24 of [10], which provides an equivalence

$$[\text{FI}\sharp, \text{Mod}^R] \simeq [\mathfrak{S}, \text{Mod}^R]$$

between the category of functors $\text{FI}\sharp \rightarrow \text{Mod}^R$ and the category of functors $\mathfrak{S} \rightarrow \text{Mod}^R$. This is the special case of (1.3) above in which $\mathcal{A}$ is the category FI of finite sets and injective functions, and $\mathcal{M}$ consists of all the morphisms. This result has provided a new viewpoint on representations of the symmetric groups, and a new viewpoint on Joyal species [20, 21].

In order to consider stability properties of representations of the symmetric groups, the authors of [10] also consider FI -modules: that is, R -module-valued functors from the category FI . Each $\text{FI}\sharp$ -module clearly has an underlying FI -module, so their Theorem 2.24 shows how symmetric group representations become FI -modules. One application they give is a structural version of the Murnaghan Theorem [31, 32], a problem which has its combinatorial aspects [37]. We are reminded of the way in which Mackey functors [28, 33] give extra freedom to representation theory.

Another instance of an equivalence of the form (1.3) is when $\mathcal{A}$ is the category of finite sets with its usual (surjective, injective)-factorization system. Then $\text{Par}\mathcal{A}$ is equivalent to the category of pointed finite sets, which is equivalent to Graeme Segal's category Γ [35]. After completing this work, we were alerted to Teimuraz Pirashvili's interesting paper [34] which gives this finite sets example, makes the connection with Dold-Puppe-Kan, and discusses stable homotopy of Γ -spaces.

We are very grateful to Aurelien Djament and Ricardo Andrade for quickly responding to our posting [arXiv:1402.7151](https://arxiv.org/abs/1402.7151) on this work by alerting us to the existence of [36] and [17]. These papers cover many of our examples and have many similar ideas. We trust our approach is complementary to these papers which, of course, we should have discovered ourselves.

Consider the basic equivalence (1.1). It is really about Cauchy (or Morita) equivalence of the free additive category on the ordinary category $\mathcal{P}$ and the free additive category on the category with zero morphisms $\mathcal{D}$. By the general theory of Cauchy completeness (see [38] for example), to have (1.1) for all Cauchy complete additive categories $\mathcal{X}$, it suffices to have it when $\mathcal{X}$ is the category of abelian groups. Cauchy completeness amounts to existence of absolute limits (see [39]) and, for additive categories, amounts to the existence of finite direct sums and splittings for idempotents.

The strategy of proof of our main equivalence (1.1) is as follows. In Section 9 we provide general criteria for deducing that a $\mathcal{V}$ -functor between $\mathcal{V}$ -functor $\mathcal{V}$ -categories is an equivalence given that some other $\mathcal{V}$ -functor between “smaller” $\mathcal{V}$ -functor $\mathcal{V}$ -categories is known to be an equivalence. In Section 2 we describe the setting for our main result and then, in Section 3 show that the partial map, Dold-Puppe-Kan and cube examples satisfy our assumptions. We show in Section 4 that the setting of Section 2 leads to an adjunction between functor categories of a particularly combinatorial kind. Section 5 reduces the proof that the adjunction is an equivalence to a particular case, then Section 6 treats that case. Section 7 provides some more examples. In the spirit of Bourn’s version [7] of Dold-Kan, Section 8 shows, under an extra assumption on our basic setting, if $\mathcal{X}$ is semiabelian then, instead of the equivalence (1.1), we have a monadic functor.

2. THE SETTING

Let $\mathcal{P}$ be a category with a subcategory $\mathcal{M}$ containing all the objects.

Assume we have a functor $(-)^*: \mathcal{M}^{\text{op}} \rightarrow \mathcal{P}$ which is the identity on objects and is such that $m^* \circ m = 1$ for all $m \in \mathcal{M}$. In particular, the morphisms in $\mathcal{M}$ are split monomorphisms (coretractions) in $\mathcal{P}$.

We write $\text{Sub}A$ for the (partially) ordered set of isomorphism classes of morphisms $m: U \rightarrow A$ into A in $\mathcal{M}$. We will use the term *subobject* rather than “ $\mathcal{M}$ -subobject” for these elements. The order on $\text{Sub}A$ is the usual one. Abusing notation for this order, we simply write $U \preceq V$ when there exists an $f: U \rightarrow V$ such that $m = n \circ f$, where $m: U \rightarrow A$ and $n: V \rightarrow A$ in $\mathcal{M}$. There can be confusion when $U = V$ as objects, so we assure the reader that we will take care. A subobject of A is *proper* when it is represented by a non-invertible $m: U \rightarrow A$ in $\mathcal{M}$; we write $U \prec A$ or $U \prec_m A$.

Define $\mathcal{R}$ to be the class of morphisms $r \in \mathcal{P}$ with the property that, if $r = m \circ x \circ n^*$ with $m, n \in \mathcal{M}$, then m, n are invertible.

Define $\mathcal{D}$ to be the category with zero morphisms (that is, 1/Set-enriched category) obtained from $\mathcal{R}$ by adjoining zero morphisms. Composition in $\mathcal{D}$ of morphisms in $\mathcal{R}$ is as in $\mathcal{P}$ if the result is itself in $\mathcal{R}$, but zero otherwise.

Define $\mathcal{S}$ to be the class of morphisms in $\mathcal{P}$ of the form $r \circ m^*$ with $m \in \mathcal{M}$ and $r \in \mathcal{R}$.

Assumption 2.1. Every morphism $f \in \mathcal{P}$ factors as $f = n \circ r \circ m^*$, uniquely up to isomorphism for $m, n \in \mathcal{M}$ and $r \in \mathcal{R}$.

Assumption 2.2. If $r, r' \in \mathcal{R}$ are composable then $r' \circ r \in \mathcal{S}$.

Assumption 2.3. If $r, m^* \circ r \in \mathcal{R}$ and $m \in \mathcal{M}$ then m is invertible.

Assumption 2.4. The class $\mathcal{M} \circ \mathcal{M}^*$ of morphisms of the form $m \circ n^*$ with $m, n \in \mathcal{M}$ is closed under composition.

Assumption 2.5. For all objects $A \in \mathcal{P}$, the ordered set $\text{Sub}A$ is finite.

Assumption 2.6. If $t \circ s = m \circ r$ with $s, t \in \mathcal{S}$, $r \in \mathcal{R}$ and $m \in \mathcal{M}$ then both s and t are in $\mathcal{R}$.

Recall that the limit of a diagram consisting of a family of morphisms into a fixed object A is called a *wide pullback*; the morphisms in the limit cone are called *projections*. The dual is *wide pushout*. The following result helps in checking examples.

Proposition 2.7. *Assume wide pullbacks of families of morphisms in $\mathcal{M}$ exist, have projections in $\mathcal{M}$, and become wide pushouts under $m \mapsto m^*$. Then Assumptions 2.1 and 2.6 hold.*

Proof. For Assumption 2.1, take $f: A \rightarrow B$ in $\mathcal{P}$. Let $n: Y \rightarrow B$ be the wide pullback of all those morphisms $V \rightarrow B$ in $\mathcal{M}$ through which f factors. Then $n \in \mathcal{M}$ and there exists a unique f_1 with $f = n \circ f_1$. Let $m: X \rightarrow A$ be the wide pullback of those $n: U \rightarrow A$ in $\mathcal{M}$ such that the morphism f_1 factors through n^* . Then $m \in \mathcal{M}$ and $f_1 = r \circ m^*$ for a unique r . Clearly $r \in \mathcal{R}$ and we have uniqueness by a familiar argument.

Next we prove a weaker form of Assumption 2.6:

if $s \circ r = m \circ r'$ with $r, r' \in \mathcal{R}$, $s \in \mathcal{S}$ and $m \in \mathcal{M}$ then $s \in \mathcal{R}$.

In obvious notation, put $s = r_1 \circ m_1^*$, $m_1^* \circ r = m_2 \circ r_2 \circ m_3^*$ and $r_1 \circ m_2 = m_4 \circ r_3 \circ m_5^*$. Then $m \circ r' = s \circ r = r_1 \circ m_1^* \circ r = r_1 \circ m_2 \circ r_2 \circ m_3^* = m_4 \circ r_3 \circ m_5^* \circ r_2 \circ m_3^*$. Let $p, q \in \mathcal{M}$ be the projections in the pullback of m, m_4 . Then there exists a unique u into the pullback with $r' = p \circ u$ and $r_3 \circ m_5^* \circ r_2 \circ m_3^* = q \circ u$. Since r' can factor through no proper $\mathcal{M}$, we deduce that p is invertible. Put $n = q \circ p^{-1}$. Then $m = m_4 \circ n$ and $n \circ r' = q \circ u = r_3 \circ m_5^* \circ r_2 \circ m_3^*$. Therefore $r' = n^* \circ r_3 \circ m_5^* \circ r_2 \circ m_3^*$; by definition of $\mathcal{R}$, we obtain that m_3 is invertible. Then $m_1^* \circ (r \circ m_3) = m_2 \circ r_2$ implies $(m_1 \circ m_2)^* \circ (r \circ m_3) = r_2$. Assumption 2.3 applies to yield that $m_1 \circ m_2$ is invertible. So m_1 has a right inverse, as well as its left inverse m_1^* , and so is invertible. So $s = r_1 \circ m_1^*$ is in $\mathcal{R}$ as asserted.

Now we come to the full Assumption 2.6. Since $s \in \mathcal{S}$, $s = r_1 \circ m_1^*$. Then, by definition of $\mathcal{R}$, $m^* \circ t \circ r_1 \circ m_1^* = m^* \circ t \circ s = m^* \circ m \circ r = r$ implies m_1 invertible. So $s \in \mathcal{R}$. Now the weaker form above applies to yield $t \in \mathcal{R}$. $\square$

For each $u: A \rightarrow B$ in $\mathcal{P}$, we have $m_u \in \mathcal{M}$ and $s_u \in \mathcal{S}$ as in the triangle

$$\begin{array}{ccc} A & \xrightarrow{u} & B \\ & \searrow s_u & \nearrow m_u \\ & S_u & \end{array} \quad (2.4)$$

by using Assumption 2.1.

Proposition 2.8. *Suppose $u: A \rightarrow B$ and $v: B \rightarrow C$ in $\mathcal{P}$.*

- (a) *If $s_{vu} \in \mathcal{R}$ then $s_u \in \mathcal{R}$.*
- (b) *If $s_{vu} \in \mathcal{R}$ and $u \in \mathcal{S}$ then $u, s_v \in \mathcal{R}$.*

Proof. We prove (b) first. We have $v \circ u = m_v \circ s_v \circ u = m_v \circ m_{s_v u} \circ s_{s_v u}$. By uniqueness of factorization, we may take $s_{vu} = s_{s_v u}$. So $s_v \circ u = m_{s_v u} \circ s_{s_v u}$. By Assumption 2.6, if $s_{vu} \in \mathcal{R}$ and $u \in \mathcal{S}$ then $s_v, u \in \mathcal{R}$. Now we prove (a). We have $v \circ u = v \circ m_u \circ s_u$. By (b), $s_{vm_u s_u} = s_{vu} \in \mathcal{R}$ implies $s_u \in \mathcal{R}$. $\square$

3. BASIC EXAMPLES

Example 3.1. Take a category $\mathcal{A}$ with a factorization system $(\mathcal{E}, \mathcal{M})$ in the sense of [15]. Assume that the pullback of any morphism with one in $\mathcal{M}$ exists. Assume every morphism in $\mathcal{M}$ is a monomorphism and every object of $\mathcal{A}$ has only finitely many $\mathcal{M}$ -subobjects. A span $f = (X \xleftarrow{f_0} U \xrightarrow{f_1} Y)$ is called a *partial map* $f: X \rightarrow Y$ when f_0 is in $\mathcal{M}$. If $g = (X \xleftarrow{g_0} V \xrightarrow{g_1} Y)$ is another partial map, we write $f \leq g$ when there exists $h: U \rightarrow V$ with $g_0 \circ h = f_0$ and $g_1 \circ h = f_1$. Let $\mathcal{P} = \text{Par}\mathcal{A}$ denote the category whose objects are all those of $\mathcal{A}$ and whose morphisms are isomorphism classes $[f]$ of partial maps. Composition is that of spans: that is, by pullback. Taking the usual order $f \leq g$ on partial maps, we can regard $\text{Par}\mathcal{A}$ as a locally partially ordered 2-category. We identify $f: X \rightarrow Y$ in $\mathcal{A}$ with the morphism $[1_X, X, f]: X \rightarrow Y$ in $\mathcal{P}$. In this way, we have the $\mathcal{M}$ we require for $\mathcal{P}$ as the one in $\mathcal{A}$. For $m: U \rightarrow X$ in $\mathcal{M}$, a right adjoint in $\text{Par}\mathcal{A}$ is defined by $m^* = [m, U, 1_U]: X \rightarrow U$, and clearly $m^* \circ m = 1$. This gives our functor $(-)^*: \mathcal{M}^{\text{op}} \rightarrow \mathcal{P}$. Every partial map $f = (X \xleftarrow{f_0} U \xrightarrow{f_1} Y)$ has

$$[f] = f_1 \circ f_0^*$$

where $f_0 \in \mathcal{M}$. Furthermore, $f_1 = m \circ e$ uniquely up to isomorphism for $m \in \mathcal{M}$ and $e \in \mathcal{E}$. It follows therefore that $\mathcal{R} = \mathcal{E}$ and that $[f] \in \mathcal{S}$ if and only if $f_1 \in \mathcal{E}$.

Now we look at our Assumptions. For Assumptions 2.1 and 2.6 we make use of Proposition 2.7. First note that since we are assuming the $\mathcal{M}$ -subobjects form a finite set, finite wide pullbacks can be obtained from pullbacks. By assumption, pullbacks of $\mathcal{M}$ s exist in $\mathcal{A}$ and a pullback of an $\mathcal{M}$ is an $\mathcal{M}$ in a factorization system. It is a pleasant exercise to see that these pullbacks remain pullbacks in $\mathcal{P}$ and become pushouts in $\mathcal{P}$ on taking left adjoints.

We certainly have Assumption 2.2; indeed $\mathcal{R}$ is closed under composition because $\mathcal{E}$ is.

For Assumption 2.3, take $r = [1_A, A, e]$ with $e \in \mathcal{E}$ and $m = [1_V, V, m]$ in $\mathcal{M}$. To have $m^* \circ r = [n, P, u] \in \mathcal{R}$, we must have n invertible and $u \in \mathcal{E}$. Then $e = m \circ u \circ n^{-1}$ implies $m \in \mathcal{E}$; so m is invertible.

The class of partial maps in Assumption 2.4 are those of the form $[m, U, n]$ with $m, n \in \mathcal{M}$; these are closed under composition since pullbacks of $\mathcal{M}$ s exist and are in $\mathcal{M}$.

Assumption 2.5 was one of our assumptions on the factorization system on $\mathcal{A}$.

Notice that $\mathcal{S}$ is not closed under composition unless each pullback of an $\mathcal{E}$ along an $\mathcal{M}$ is an $\mathcal{E}$. This is true in many examples.

Example 3.2. Take $\mathcal{P}$ to be the category $\Delta_{\perp, \top}$ of finite non-empty ordinals $n = \{0, 1, \dots, n-1\}$ with morphisms those functions which preserve first element, last element and order. Functors out of this category are augmented simplicial objects because of the isomorphism (1.2). Morphisms $\xi, \zeta: m \rightarrow n$ are ordered by taking $\xi \leq \zeta$ iff and only if $\xi(i) \leq \zeta(i)$ for all $i \in m$. This makes $\Delta_{\perp, \top}$ into a locally partially

ordered 2-category; it is a locally full sub-2-category of Cat . Take $\mathcal{M}$ to consist of all the injective functions in $\Delta_{\perp, \top}$; each such injection ∂ has a left adjoint ∂^* (and also a right adjoint for that matter) in $\Delta_{\perp, \top}$; clearly ∂^* is surjective with $\partial^* \circ \partial = 1$ since ∂ is a fully faithful functor. A surjection σ in $\mathcal{P}$ is of the form ∂^* if and only if $\sigma(i) = 0$ implies $i = 0$. We write $\sigma_k: m+1 \rightarrow m$ for the order-preserving surjection which takes the value k twice. We write $\partial_k: m \rightarrow m+1$ for the order-preserving injection which does not have k in its image. Note that $\partial_0 \notin \mathcal{P}$, while $\sigma_k \dashv \partial_k \dashv \sigma_{k-1}$ in $\mathbb{P}$ and σ_k is a ∂^* if and only if $k > 0$. Every $\xi \in \mathcal{P}$ factors uniquely as

$$\xi = \partial_{i_s} \circ \cdots \circ \partial_{i_1} \circ \sigma_{j_1} \circ \cdots \circ \sigma_{j_t}$$

for $0 < i_1 < \cdots < i_s < n-1$ and $0 \leq j_1 < \cdots < j_t < m-1$.

We claim $\mathcal{R}$ consists of the identities and the surjections $\sigma_0: m+1 \rightarrow m$. The only invertible morphisms in $\mathcal{P}$ are identities. Since members of $\mathcal{R}$ factor through no proper injection, they must be surjective. Every surjection τ is either of the form ∂^* or uniquely of the form $\sigma_0 \circ \partial^*$. Neither of these forms is permissible for $\tau \in \mathcal{R}$ unless the injection ∂ is an identity. This proves our claim. It is also clear then that $\mathcal{S}$ consists of all the surjections in $\mathcal{P}$.

To prove the Assumptions, we make use of the simplicial identities (see page 24 of [16] for example) which, apart from $\sigma_i \circ \partial_i = 1 = \sigma_{i-1} \circ \partial_i$, say that, for all $i < j$,

$$\partial_j \circ \partial_i = \partial_i \circ \partial_{j-1}, \quad \sigma_{j-1} \circ \sigma_i = \sigma_i \circ \sigma_j, \quad \sigma_j \circ \partial_i = \partial_i \circ \sigma_{j-1}, \quad \sigma_i \circ \partial_{j+1} = \partial_j \circ \sigma_i.$$

The hypotheses of Proposition 2.7 hold since the pullback of any two monomorphisms in $\mathcal{P}$ exists and is absolute (that is, preserved by all functors); see page 27 of [16] on the Eilenberg-Zilber Theorem. Alternatively, notice that the squares

$$\begin{array}{ccc} n-1 & \xrightarrow{\partial_{j-1}} & n \\ \partial_i \downarrow & & \downarrow \partial_i \\ n & \xrightarrow{\partial_j} & n+1 \end{array} \quad \begin{array}{ccc} n-1 & \xleftarrow{\sigma_{j-1}} & n \\ \sigma_i \uparrow & & \uparrow \sigma_i \\ n & \xleftarrow{\sigma_j} & n+1 \end{array}$$

are respectively a pullback and pushout in $\mathcal{P}$ for $0 < i < j < n$. Then the general result follows by stacking these squares vertically and horizontally.

In the present example, $\mathcal{S}$ is closed under composition, making Assumption 2.2 clear.

For Assumption 2.3, suppose we have $\partial \circ \rho \in \mathcal{R}$ and $\rho \in \mathcal{R}$. Since ρ is surjective, $\partial^* \circ \rho = 1$ implies $\rho = 1$ and hence $\partial = 1$, as required. Otherwise $\partial^* \circ \rho = \sigma_0$. Yet, if $\rho = 1$ this contradicts the lack of right adjoint for σ_0 . So $\rho = \sigma_0$. Thus $\partial^* \circ \sigma_0 = \sigma_0$, and we can cancel σ_0 to obtain again $\partial = 1$.

For Assumption 2.4, the class $\mathcal{M} \circ \mathcal{M}^*$ of morphisms consists of those which reflect 0. That is, $\xi = \mu \circ \partial^*$ with $\mu \in \mathcal{M}$ if and only if $\xi(i) = 0$ implies $i = 0$. This class is clearly closed under composition.

Finally, Assumption 2.5 is clear.

Notice that the arguments above equally apply to the full subcategory $\Delta_{\perp \neq \top}$ of $\Delta_{\perp, \top}$ obtained by removing the object 1. Functors $\Delta_{\perp \neq \top} \rightarrow \mathcal{X}$ are the traditional simplicial objects in $\mathcal{X}$.

Example 3.3. This example is about the cubical category $\mathbb{I}$ as used by Sjoerd Crans [11] and Dominic Verity [40, 41]. Functors with domain $\mathbb{I}$ are cubical objects in the

codomain category. Verity constructed $\mathbb{I}$ as the free monoidal category containing a cointerval.

For each natural number k , define a poset $\langle k \rangle = \{-, 1, 2, \dots, k, +\}$ by adjoining a bottom element $-$ and a top element $+$ to the discrete poset $\{1, 2, \dots, k\}$. Any function $f: \langle k \rangle \rightarrow \langle h \rangle$ which preserves top and bottom is order-preserving. Thus we get a locally partially ordered 2-category with objects the $\langle k \rangle$, with morphisms the top-and-bottom-preserving functions, and with the pointwise order. Let $\mathbb{I}$ be the locally full sub-2-category consisting of those $f: \langle k \rangle \rightarrow \langle h \rangle$ for which, if $f(i), f(j) \notin \{-, +\}$ then $i < j$ if and only if $f(i) < f(j)$.

Let $\mathcal{P}$ be the underlying category of this $\mathbb{I}$. Let $\mathcal{M}$ consist of the morphisms in $\mathcal{P}$ which are injective as functions. Given such an $m: \langle k \rangle \rightarrow \langle h \rangle$ in $\mathcal{M}$, define $m^*: \langle h \rangle \rightarrow \langle k \rangle$ to send each $m(i)$ in the image of m to i and everything else to $+$. Clearly $m^* \in \mathcal{P}$ and $m^* \circ m = 1$. Furthermore, $mm^*(j)$ is equal to j if $j = m(i)$ for some i , and $+$ otherwise. Therefore $1 \leq m \circ m^*$ showing m^* to be left adjoint to m with identity counit.

We can characterize morphisms of the form m^* as those which are surjective as functions and reflect the bottom element $-$. Consequently $\mathcal{R}$ consists of the morphisms which are surjective as functions and reflect the top element $+$.

Assumption 2.5 and 2.2 are clear.

$$\begin{array}{ccc}
 \langle \ell \rangle & \xrightarrow{q} & \langle v \rangle \\
 p \downarrow & & \downarrow n \\
 \langle u \rangle & \xrightarrow{m} & \langle k \rangle
 \end{array}
 \qquad
 \begin{array}{ccc}
 \langle k \rangle & \xrightarrow{n^*} & \langle v \rangle \\
 m^* \downarrow & & \downarrow q^* \\
 \langle u \rangle & \xrightarrow{p^*} & \langle \ell \rangle
 \end{array}
 \begin{array}{c}
 \searrow g \\
 \downarrow f \\
 \langle h \rangle
 \end{array}
 \tag{3.5}$$

For Assumptions 2.1 and 2.6, the existence of intersections is obvious. However, we must show that taking left adjoints gives cointersections. Take a pullback as in the left-hand diagram of (3.5) and consider the right-hand diagram. Assume $f \circ m^* = g \circ n^*$. It suffices to show $f \circ p \circ p^* = f$. Now $fpp^*(i)$ is equal to $fp(j)$ if $i = p(j)$ for some j , and equal to $+$ otherwise. In the first case, we have $fpp^*(i) = fp(j) = f(i)$, as required. In the second case, if i does not have the form $p(j)$ then $m(i)$ is not in the image of n , and so $gn^*m(i) = +$; thus $f(i) = fm^*m(i) = +$ and we again have $fpp^*(i) = f(i)$.

For Assumption 2.3, suppose $m \in \mathcal{M}$ and $r, m^* \circ r \in \mathcal{R}$. Suppose $m^*(i) = +$. Since r is surjective, we have $i = r(j)$ so that $m^*r(j) = +$. However $m^* \circ r$ reflects $+$. So $j = +$, yielding $i = r(+) = +$. This proves m^* reflects $+$ and therefore must be invertible.

For Assumption 2.4, the composites of the form $m \circ n^*$ are clearly the (not necessarily surjective) morphisms which reflect $-$. Clearly these are closed under composition.

There is another possible characterization of $\mathcal{R}$, namely as the category of right adjoints to the morphisms in $\mathcal{M}$. Thus in fact $\mathcal{R}$ is dual to $\mathcal{M}$. Now $\mathcal{M}$ is really just the category Δ_{inj} of finite ordinals and injective order-preserving maps, and $\mathcal{R} \cong \mathcal{M}^{\text{op}}$. Also the category $\mathcal{M}^*$, with morphisms the $m^* \in \mathcal{M}$, is dual to $\mathcal{M}$. The factorization

of Proposition 2.7 in this case shows the category $\mathcal{P}$ is a composite

$$\mathbb{I} = \Delta_{\text{inj}} \circ \Delta_{\text{inj}}^{\text{op}} \circ \Delta_{\text{inj}}^{\text{op}},$$

relative to suitably defined distributive laws.

An alternative viewpoint is that $\mathbb{I}$ is $\text{ParPar}\Delta_{\text{inj}}$, where in each case partial maps are defined relative to the morphisms in Δ_{inj} .

4. THE KERNEL MODULE

We write $1/\text{Set}$ for the category of pointed sets. We write $X \wedge Y$ for the monoidal tensor on pointed sets. Let $\text{p}\Lambda$ denote the free pointed set $1 + \Lambda$ on the set Λ . This defines the value on objects of a strong monoidal functor $\text{p}: \text{Set} \rightarrow 1/\text{Set}$ whose right adjoint is the forgetful functor.

In the setting of Section 2, we shall define a functor

$$M: \mathcal{D}^{\text{op}} \times \mathcal{P} \longrightarrow 1/\text{Set} \quad (4.6)$$

which preserves zeros in the first variable.

Using the notation of (2.4), define M on objects by

$$M(A, B) = \text{p}\{u \in \mathcal{P}(A, B) : s_u \in \mathcal{R}\}. \quad (4.7)$$

Suppose $r: A_1 \rightarrow A$ in $\mathcal{R}$ and $f: B \rightarrow B_1$ in $\mathcal{P}$. Define M on morphisms by

$$M(r, f)u = \begin{cases} f \circ u \circ r & \text{for } s_{f_{ur}} \in \mathcal{R} \\ 0 & \text{otherwise.} \end{cases} \quad (4.8)$$

To verify that this is a functor we need to see that

$$M(r_1, f_1)M(r, f)u = M(r \circ_{\mathcal{D}} r_1, f_1 \circ f)u$$

holds. This amounts to showing that the left-hand side is non-zero if and only if the right-hand side is, because then both sides equal $f_1 \circ f \circ u \circ r \circ r_1$. In other words, we need to see that $s_{f_{ur}}, s_{f_1 f_{urr_1}} \in \mathcal{R}$ if and only if $r \circ r_1, s_{f_1 f_{urr_1}} \in \mathcal{R}$. In fact, $s_{f_1 f_{urr_1}} \in \mathcal{R}$ implies both $s_{f_{ur}} \in \mathcal{R}$ and $r \circ r_1 \in \mathcal{R}$. For we know by Assumption 2.2 that $r \circ r_1 \in \mathcal{S}$; so, by Proposition 2.8(b), $s_{f_1 f_{urr_1}} \in \mathcal{R}$ implies $r \circ r_1 \in \mathcal{R}$. By Proposition 2.8(a), we also conclude that $s_{f_{urr_1}} \in \mathcal{R}$; then Proposition 2.8(b) gives $s_{f_{ur}} \in \mathcal{R}$.

Let $\text{p}_*\mathcal{P}$ denote the free $1/\text{Set}$ -enriched category on the category $\mathcal{P}$. For any locally pointed (that is, $1/\text{Set}$ -enriched) category $\mathcal{X}$, we have an isomorphism of categories

$$[\text{p}_*\mathcal{P}, \mathcal{X}]_{\text{pt}} \cong [\mathcal{P}, \mathcal{X}],$$

where, for emphasis, we write the subscript “pt” for the pointed-set-enriched functor category.

We identify (4.6) with its obvious extension to a $1/\text{Set}$ -functor

$$M: \mathcal{D}^{\text{op}} \otimes \text{p}_*\mathcal{P} \longrightarrow 1/\text{Set}. \quad (4.9)$$

Now we are in the situation for a kernel adjunction of the form (9.29) with $\mathcal{V} = 1/\text{Set}$. For any suitably complete and cocomplete $1/\text{Set}$ -category $\mathcal{X}$, we have an adjunction

$$[\mathcal{D}, \mathcal{X}]_{\text{pt}} \begin{array}{c} \xrightarrow{\widehat{M}} \\ \perp \\ \xleftarrow{\widetilde{M}} \end{array} [\mathcal{P}, \mathcal{X}]. \quad (4.10)$$

More combinatorial expressions than (9.31) and (9.30) exist for $\widehat{M}$ and $\widetilde{M}$ in our present situation.

Theorem 4.1. *Suppose M is the kernel as in (4.9). Let $\mathcal{X}$ be any suitably complete and cocomplete $1/\text{Set}$ -category. For any zero preserving functor $F: \mathcal{D} \rightarrow \mathcal{X}$ and any functor $T: \mathcal{P} \rightarrow \mathcal{X}$, there are isomorphisms*

$$\widehat{M}(F)B \cong \sum_{S \preceq_m B} FS, \quad \widetilde{M}(T)A \cong \bigcap_{U \prec_m A} \ker T(m^*: A \rightarrow U).$$

Proof. We shall prove

$$\int^{A \in \mathbb{Z}_+ \mathcal{D}} M(A, B) \wedge FA \cong \sum_{S \preceq_m B} FS$$

by showing that the family of morphisms χ_A as defined by the diagram

$$\begin{array}{ccc} M(A, B) \wedge FA & \xrightarrow{\chi_A} & \sum_{S \preceq_m B} FS \\ \text{in}_u \uparrow & & \uparrow \text{in}_{m_u} \\ FA & \xrightarrow{Fs_u} & FS_u, \end{array} \quad (4.11)$$

for $u \in M(A, B)$, using the notation of (2.4), is a universal extraordinary natural transformation (that is, a universal wedge). In order for a family $\theta_A: M(A, B) \wedge FA \rightarrow X$ in $\mathcal{X}$ to be extraordinary natural, we require

$$\theta_A \circ \text{in}_u \circ Fr = \begin{cases} \theta_{A_1} \circ \text{in}_{ur} & \text{for } s_{ur} \in \mathcal{R} \\ 0 & \text{otherwise.} \end{cases} \quad (4.12)$$

for all $r: A_1 \rightarrow A$ in $\mathcal{R}$ and $u: A \rightarrow B$ in $\mathcal{P}$ with $s_u \in \mathcal{R}$. Notice that in this situation $s_{ur} = s_u \circ r$ by uniqueness of factorization and Assumption 2.2.

Now for the family χ we have $\chi_A \circ \text{in}_u \circ Fr = \text{in}_{m_u} \circ F(s_u \circ_{\mathcal{D}} r)$. If $s_{ur} \in \mathcal{R}$ then $s_u \circ_{\mathcal{D}} r = s_u \circ r = s_{ur}$, so $\text{in}_{m_u} \circ F(s_u \circ_{\mathcal{D}} r) = \text{in}_{m_u} \circ Fs_{ur} = \chi_{A_1} \circ \text{in}_{ur}$; otherwise, $F(s_u \circ_{\mathcal{D}} r) = F(0) = 0$, and we have our extraordinary naturality.

For universality, we must see that a general wedge θ factors uniquely through χ . Define $\phi: \sum_{S \preceq_m B} FS \rightarrow X$ by $\phi \circ \text{in}_m = \theta_S \circ \text{in}_m$. With $s_u \in \mathcal{R}$, we have, using (4.12),

$$\theta_A \circ \text{in}_u = \theta_A \circ \text{in}_{m_u s_u} = \theta_{s_u} \circ \text{in}_{m_u} \circ Fs_u = \phi \circ \text{in}_{m_u} \circ Fs_u = \phi \circ \chi_A \circ \text{in}_u.$$

Thus $\theta = \phi \chi$. By taking $u = m \in \mathcal{M}$, we also see that the definition of ϕ is forced.

Define $\widetilde{T}A = \bigcap_{U \prec_m A} \ker T(m^*: A \rightarrow U)$ with inclusion $\iota_A: \widetilde{T}A \rightarrow TA$, and, for $r: A \rightarrow A_1$, define $\widetilde{T}r: \widetilde{T}A \rightarrow \widetilde{T}A_1$ to be the restriction of Tr . For this we need to see that $Tr \circ \iota_A$ factors through ι_{A_1} . Take a non-invertible $n: V \rightarrow A_1$. Using Proposition 2.7, we have $n^* \circ r = \ell \circ r_1 \circ m^*$. So $(n \circ \ell)^* \circ r = r_1 \circ m^*$. If m is invertible then $(n \circ \ell)^* \circ r \in \mathcal{R}$. By Assumption 2.3, $n \circ \ell$ is invertible. So n is invertible, a contradiction. So m is not invertible and we have $Tn^* \circ Tr \circ \iota_A = T(\ell \circ r_1) \circ Tm^* \circ \iota_A = T(\ell \circ r_1) \circ 0 = 0$, yielding $\widetilde{T}r$ with $\iota_{A_1} \circ \widetilde{T}r = Tr \circ \iota_A$.

Put $\widehat{F}B = \sum_{S \preceq_m B} FS$ and transport the functoriality in B across the isomorphism with $\widehat{M}(F)B$. We will prove that there is a natural isomorphism

$$[\mathcal{P}, \mathcal{X}](\widehat{F}, T) \cong [\mathcal{D}, \mathcal{X}]_{\text{pt}}(F, \widetilde{T})$$

and hence conclude that $\widetilde{M}(T) \cong \widetilde{T}$. Take a natural transformation $\theta: \widehat{F} \Rightarrow T: \mathcal{P} \rightarrow \mathcal{X}$. The definition of $\widehat{F}f$ involves writing $f \circ m = m_{fm} \circ s_{fm}$ then, with $s_{fm} \in \mathcal{R}$, we have commutativity in the following diagram.

$$\begin{array}{ccccc}
 FU & \xrightarrow{\text{in}_U} & \widehat{F}A & \xrightarrow{\theta_A} & TA \\
 \downarrow F s_{fm} & & \downarrow \widehat{F}f & & \downarrow Tf \\
 FV & \xrightarrow{\text{in}_V} & \widehat{F}B & \xrightarrow{\theta_B} & TB
 \end{array} \tag{4.13}$$

Consider a noninvertible $\ell: W \rightarrow A$ in $\mathcal{M}$. Then, by the properties of $\mathcal{R}$, $\ell^* \in \mathcal{S}$ and $\ell^* \notin \mathcal{R}$. So $T\ell^* \circ \theta_A \circ \text{in}_A = \theta_W \circ \widehat{F}\ell^* \circ \text{in}_A = \theta_W \circ 0 = 0$. This implies there exists a unique morphism $\phi_A: FA \rightarrow \widetilde{T}A$ such that the following square commutes.

$$\begin{array}{ccc}
 FA & \xrightarrow{\phi_A} & \widetilde{T}A \\
 \downarrow \text{in}_A & & \downarrow i_A \\
 \widehat{F}A & \xrightarrow{\theta_A} & TA
 \end{array} \tag{4.14}$$

Naturality of ϕ is proved as follows using (4.13) with $f = r \in \mathcal{R}$: $i_B \circ \phi_B \circ Fr = \theta_B \circ \text{in}_B \circ Fr = \theta_B \circ \widehat{F}r \circ \text{in}_A = Tr \circ \theta_A \circ \text{in}_A = Tr \circ i_A \circ \phi_A = i_B \circ \widetilde{T}r \circ \phi_A$.

For the inverse direction, take any natural transformation $\phi: F \Rightarrow \widetilde{T}: \mathcal{D} \rightarrow \mathcal{X}$. Define θ by commutativity of the following diagram.

$$\begin{array}{ccccc}
 FU & \xrightarrow{\text{in}_U} & \widehat{F}A & \xrightarrow{\theta_A} & TA \\
 \downarrow \phi_U & & & & \uparrow Tm \\
 \widetilde{T}U & \xrightarrow{i_U} & TU & &
 \end{array} \tag{4.15}$$

We need to prove the right-hand square of (4.13) commutes when precomposed with any in_U for any $m: U \rightarrow A$ in $\mathcal{M}$. In the case where $s_{fm} \in \mathcal{R}$, the desired commutativity is a consequence of the commutativity of the following three squares.

$$\begin{array}{ccccccc}
 FU & \xrightarrow{\phi_U} & \widetilde{T}U & \xrightarrow{i_A} & TU & \xrightarrow{Tm} & TA \\
 \downarrow F s_{fm} & & \downarrow \widetilde{T} s_{fm} & & \downarrow T s_{fm} & & \downarrow Tf \\
 FV & \xrightarrow{\phi_V} & \widetilde{T}V & \xrightarrow{i_B} & TV & \xrightarrow{T n_{fm}} & TB
 \end{array} \tag{4.16}$$

In the case where $s_{fm} \notin \mathcal{R}$, we can write $s_{fm} = r \circ \ell^*$ for some noninvertible $\ell \in \mathcal{M}$. Then (using both $U \preceq A$ and $U \preceq U$) we have $Tf \circ \theta_A \circ \text{in}_U = T(f \circ m) \circ i_U \circ \phi_U = T(n_{fm} \circ r) \circ T\ell^* \circ i_U \circ \phi_U = T(n_{fm} \circ r) \circ 0 \circ \phi_U = 0 = \theta_B \circ \widehat{F}f \circ \text{in}_U$, as required.

To show that the assignments are mutually inverse, take θ and define ϕ by (4.14). Let $\bar{\theta}$ be as θ is in (4.15). Then $\bar{\theta}_A \circ \text{in}_U = Tm \circ i_U \circ \phi_U = Tm \circ \theta_U \circ \text{in}_U = \theta_A \circ \widehat{F}m \circ \text{in}_U = \theta_A \circ \text{in}_U$. So $\bar{\theta} = \theta$.

On the other hand, take ϕ and define θ by (4.15). Let ϕ' be as ϕ is in (4.14). Then $i_A \circ \phi'_A = \theta_A \circ \text{in}_A = i_A \circ \phi_A$. So $\phi' = \phi$. $\square$

5. REDUCTION OF THE PROBLEM

The problem being referred to in the title of this section is to show that the adjunction (4.10) is an equivalence. We wish to use the theory in Section 9 to reduce the problem to the special case where $\mathcal{R}$ consists only of invertible morphisms. In Section 6, we will treat this special case.

Consider $\mathcal{P}$, $\mathcal{M}$, $\mathcal{I}$, $\mathcal{R}$ and $\mathcal{D}$ as in the setting of Section 2.

Let $\mathcal{I}$ be the class of invertible morphisms in $\mathcal{P}$. Let $\mathcal{K} = \mathcal{M} \circ \mathcal{M}^*$ be the subcategory of $\mathcal{P}$ as assured by Assumption 2.4. Then $\mathcal{K}$, $\mathcal{M}$, $\mathcal{M}^*$, $\mathcal{I}$ and $p_*\mathcal{I}$ also fit the setting of Section 2. Denote the kernel module for this situation by $N: \mathcal{I}^{\text{op}} \times \mathcal{K} \rightarrow 1/\text{Set}$; on objects it is given by $N(C, D) = p_*\mathcal{M}(C, D)$.

Let $H: \mathcal{I} \rightarrow \mathcal{D}$ and $K: \mathcal{K} \rightarrow \mathcal{P}$ be the inclusions; both are the identity on objects and so Cauchy dense.

We have the obvious inclusion

$$\theta_{C,D}: N(C, D) \rightarrow M(HC, KD) \quad (5.17)$$

taking $m \in \mathcal{M}(C, D)$ to $m \in \mathcal{P}(C, D)$ (using $s_m = 1$).

Proposition 5.1. *For $\theta: N \Rightarrow M(H, K)$ defined at (5.17), the corresponding natural families*

$$\begin{aligned} \theta_{A,D}^r: \int^{C \in \mathcal{I}} N(C, D) \wedge \mathcal{D}(A, HC) &\longrightarrow M(A, KD), \\ \theta_{C,B}^\ell: \int^{D \in \mathcal{K}} p_*\mathcal{P}(KD, B) \wedge N(C, D) &\longrightarrow M(HC, B), \end{aligned}$$

as in Section 9, are both invertible.

Proof. For $r: A \rightarrow C$ in $\mathcal{R}$ and $m: C \rightarrow D$ in $\mathcal{M}$, we have $\theta_{A,D}^r[m \wedge r] = m \circ r$. Yet every $u: A \rightarrow D$ with $s_u \in \mathcal{R}$ factors uniquely up to a morphism in $\mathcal{I}$ as $u = m_u \circ s_u = \theta_{A,D}^r[m_u \wedge s_u]$. So we have our inverse bijection.

By the definition of the second coend, for all $k: D \rightarrow D_1$ in $\mathcal{K}$ and $g: D_1 \rightarrow B$, we have $[(g \circ k) \wedge m] = [(g \wedge k) \circ m]$ when $k \circ m \in \mathcal{M}$, zero otherwise. For $m: C \rightarrow D$ in $\mathcal{M}$ and $f: D \rightarrow B$ in $\mathcal{P}$, we have $[f \wedge m] = [(f \circ m) \wedge 1] = [u \wedge 1]$ for some $u \in \mathcal{P}$. But $u = \ell \circ r \circ n^*$ for some $n, \ell \in \mathcal{M}$ and $r \in \mathcal{R}$. So $[u \wedge 1] = [(\ell \circ r) \wedge n^*]$ provided $n^* \in \mathcal{M}$, zero otherwise. However, $n^* \in \mathcal{M}$ implies n invertible. So $s_u \in \mathcal{R}$. In other words, $[f \wedge m]$ is either zero or of the form $[u \wedge 1]$ for a unique u with $s_u \in \mathcal{R}$. By definition, $\theta_{C,B}^\ell[f \wedge m] = f \circ m$ when $s_{fm} \in \mathcal{R}$, zero otherwise. We have shown that an inverse to $\theta_{C,B}^\ell$ is provided by $u \mapsto [u \wedge 1_C]$. $\square$

Combining Proposition 5.1 and Corollary 9.3, we have:

Corollary 5.2. *If $\tilde{N}$ is an equivalence or crudely monadic then so is $\tilde{M}$.*

6. THE CASE $\mathcal{P} = \mathcal{K}$

Let $\mathcal{K}$ be a category with a subcategory $\mathcal{M}$ which contains all the isomorphisms. Assume we have a functor $(-)^*: \mathcal{M}^{\text{op}} \rightarrow \mathcal{K}$ such that $m^* \circ m = 1$ for all $m \in \mathcal{M}$. Assume every morphism f in $\mathcal{K}$ factors as $f = m \circ \ell^*$ with $m, \ell \in \mathcal{M}$. Then $(\mathcal{M}^*, \mathcal{M})$ is a factorization system on $\mathcal{K}$ in the sense of [15]; indeed, $\mathcal{M}^*$ consists

of the retractions and $\mathcal{M}$ of the coretractions. We also assume the set $\text{Sub}A$ of $\mathcal{M}$ -subobjects of each object A is finite. Let $\mathcal{I}$ be the groupoid of invertible morphisms in $\mathcal{K}$.

Let $J: \mathcal{I} \rightarrow \mathcal{M}$ and $I: \mathcal{M} \rightarrow \mathcal{K}$ be the inclusion functors. Both functors are the identity on objects.

Let $\mathcal{X}$ be any category admitting finite coproducts and finite products.

Lemma 6.1. *Each functor $F: \mathcal{I} \rightarrow \mathcal{X}$ has a pointwise left Kan extension*

$$\begin{array}{ccc} \mathcal{I} & \xrightarrow{J} & \mathcal{M} \\ & \searrow F & \swarrow \text{Lan}_J F \\ & \mathcal{X} & \end{array} \quad \begin{array}{c} \xRightarrow{\kappa} \\ \end{array} \quad (6.18)$$

along $J: \mathcal{I} \rightarrow \mathcal{M}$ defined on objects by:

$$(\text{Lan}_J F)X = \sum_{U \preceq_m X} FU .$$

For morphisms $f: X \rightarrow Y$ in $\mathcal{M}$, the following triangle commutes.

$$\begin{array}{ccc} FU & \xrightarrow{\text{in}_m} & (\text{Lan}_J F)X \\ & \searrow \text{in}_{fm} & \downarrow (\text{Lan}_J F)f \\ & & (\text{Lan}_J F)Y \end{array} \quad (6.19)$$

Proof. The inclusion $U \preceq_m X \mapsto (U, m: JU \rightarrow X)$ of the discrete category on the set $\{U: U \preceq_m X\}$ into the comma category J/X has a left adjoint, taking $(V, f: JV \rightarrow X)$ to $V \preceq_f X$. It is an adjoint equivalence. It follows that the colimit of $J/X \xrightarrow{\text{dom}} \mathcal{I} \xrightarrow{F} \mathcal{X}$ can be calculated by restricting along the inclusion and so is the coproduct displayed. $\square$

Lemma 6.2. *Each functor $T: \mathcal{M} \rightarrow \mathcal{X}$ has a pointwise right Kan extension*

$$\begin{array}{ccc} \mathcal{M} & \xrightarrow{I} & \mathcal{K} \\ & \searrow T & \swarrow \text{Ran}_I T \\ & \mathcal{X} & \end{array} \quad \begin{array}{c} \xleftarrow{\rho} \\ \end{array} \quad (6.20)$$

along $I: \mathcal{M} \rightarrow \mathcal{K}$ defined on objects by:

$$(\text{Ran}_I T)X = \prod_{U \preceq_m X} TU .$$

For morphisms $k: X \rightarrow Y$ in $\mathcal{K}$, the square

$$\begin{array}{ccc} (\text{Ran}_I T)X & \xrightarrow{\text{pr}_m} & TU \\ (\text{Ran}_I T)k \downarrow & & \downarrow T\ell \\ (\text{Ran}_I T)Y & \xrightarrow{\text{pr}_n} & TV \end{array} \quad (6.21)$$

commutes, where $n^* \circ k = \ell \circ m^*$ with $n, m, \ell \in \mathcal{M}$.

Proof. The functor $U \preceq_m X \mapsto (m^*: X \rightarrow IU, U)$ from the discrete category on the set $\{U: U \preceq_m X\}$ into the comma category X/I has a right adjoint, taking $(n \circ \ell^*: X \rightarrow IA, A)$ to $W \preceq_n X$, and so is initial. It follows that the limit of $X/I \xrightarrow{\text{cod}} \mathcal{M} \xrightarrow{T} \mathcal{X}$ can be calculated as the product displayed. $\square$

This gives the two adjunctions

$$[\mathcal{I}, \mathcal{X}] \begin{array}{c} \xrightarrow{\text{Lan}_J} \\ \perp \\ \xleftarrow{[J,1]} \end{array} [\mathcal{M}, \mathcal{X}] \begin{array}{c} \xrightarrow{\text{Ran}_I} \\ \top \\ \xleftarrow{[I,1]} \end{array} [\mathcal{K}, \mathcal{X}]. \quad (6.22)$$

Each adjunction generates a comonad on $[\mathcal{M}, \mathcal{X}]$:

$$G = \text{Lan}_J \circ [J, 1] \quad \text{and} \quad H = [I, 1] \circ \text{Ran}_I. \quad (6.23)$$

To be more explicit, the functors GF and HF are defined on objects by

$$(GF)A = \sum_{U \preceq A} FU \quad \text{and} \quad (HF)A = \prod_{U \preceq A} FU.$$

On morphisms $f: A \rightarrow B$ in $\mathcal{M}$, they are defined by commutativity in the diagrams

$$\begin{array}{ccc} & FU & \\ \text{in}_m \swarrow & & \searrow \text{in}_{fm} \\ (GF)A & \xrightarrow{(GF)f} & (GF)B \end{array} \quad \begin{array}{ccc} (HF)A & \xrightarrow{(HF)f} & (HF)B \\ \text{pr}_m \downarrow & & \downarrow \text{pr}_n \\ FU & \xrightarrow{F\ell} & FV \end{array}, \quad (6.24)$$

where $n^* \circ f = \ell \circ m^*$ with $n, m, \ell \in \mathcal{M}$. The counits $\varepsilon_F: GF \rightarrow F$ and $\varepsilon_F: HF \rightarrow F$ have components respectively defined by $\varepsilon_F A \circ \text{in}_m = Fm$ and $\varepsilon_F A = \text{pr}_{1_A}$. The comultiplications $\delta_F: GF \rightarrow G^2F$ and $\delta_F: HF \rightarrow H^2F$ have components respectively defined as follows.

$$\begin{array}{ccc} FU & \xrightarrow{\text{in}_m} & \sum_{U \preceq_m A} FU \\ & \searrow \text{in}_{1_U, m} & \swarrow \delta_F A \\ & \sum_{V \preceq_\ell U \preceq_m A} FV & \\ \prod_{U \preceq_m A} FU & \xrightarrow{\text{pr}_{m\ell}} & FV \\ & \searrow \delta_F A & \swarrow \text{pr}_{\ell, m} \\ & \prod_{V \preceq_\ell U \preceq_m A} FV & \end{array}$$

We shall construct a comonad morphism $\Theta: G \Rightarrow H$ when $\mathcal{X}$ is a $1/\text{Set}$ -category. Take $F: \mathcal{M} \rightarrow \mathcal{X}$ and $A \in \mathcal{M}$. We define

$$\Theta_F A: \sum_{U \preceq_m A} FU \longrightarrow \prod_{V \preceq_n A} FV \quad (6.25)$$

by taking the composite

$$FU \xrightarrow{\text{in}_m} \sum_{U \preceq_m A} FU \xrightarrow{\Theta_F A} \prod_{V \preceq_n A} FV \xrightarrow{\text{pr}_n} FV$$

to be zero unless $n^* \circ m \in \mathcal{M}$, in which case the composite is $F(n^* \circ m): FU \rightarrow FV$.

Proposition 6.3. *For the two comonads G and H of (6.23) on $[\mathcal{M}, \mathcal{X}]$, a comonad morphism $\Theta: G \Rightarrow H$ is defined by the components (6.25).*

Proof. Naturality in A is proved by contemplating the following diagram in which $n^* \circ f = \ell_1 \circ m_1^*$.

$$\begin{array}{ccccccc}
 FU & \xrightarrow{\text{in}_m} & \sum_{U \preceq_m A} FU & \xrightarrow{\Theta_F A} & \prod_{U_1 \preceq_{m_1} A} FU_1 & \xrightarrow{\text{pr}_{m_1}} & FU_1 \\
 & \searrow \text{in}_{fm} & \downarrow (GF)f & & \downarrow (HF)f & & \downarrow F\ell_1 \\
 & & \sum_{V_1 \preceq_{n_1} B} FV_1 & \xrightarrow{\Theta_F B} & \prod_{V \preceq_n B} FV & \xrightarrow{\text{pr}_n} & FV
 \end{array}$$

The top composite is zero unless $m_1^* \circ m \in \mathcal{M}$. The bottom composite is zero unless $n^* \circ f \circ m \in \mathcal{M}$. Notice that $\ell_1 \circ m_1^* \circ m = n^* \circ f \circ m \in \mathcal{M}$ if and only if $m_1^* \circ m \in \mathcal{M}$. So the top composite is non-zero exactly when the bottom composite is; in that case, the composites are equal since $F\ell_1 \circ F(m_1^* \circ m) = F(n^* \circ f \circ m)$.

Naturality in F is obvious.

Preservation of the counits is proved by the calculation

$$\varepsilon_F A \circ \Theta_F A \circ \text{in}_m = \text{pr}_A \circ \Theta_F A \circ \text{in}_m = Fm = \varepsilon_F A \circ \text{in}_m .$$

Preservation of the comultiplications is proved by observing that

$$\text{pr}_{\ell, m} \circ \delta_F A \circ \Theta_F A \circ \text{in}_{m_1} = \text{pr}_{m\ell} \circ \Theta_F A \circ \text{in}_{m_1}$$

while

$$\text{pr}_{\ell, m} \circ \Theta_F^2 A \circ \delta_F A \circ \text{in}_{m_1} = \text{pr}_{\ell, m} \circ \Theta_F^2 A \circ \text{in}_{1_{U_1}, m_1} .$$

Both of these right-hand sides are zero unless $\ell^* \circ m^* \circ m_1 \in \mathcal{M}$ (noting this implies $m^* \circ m_1 \in \mathcal{M}$), in which case they are both equal to $F(\ell^* \circ m^* \circ m_1)$. $\square$

Proposition 6.4. *The functor $[I, 1]: [\mathcal{K}, \mathcal{X}] \rightarrow [\mathcal{M}, \mathcal{X}]$ is comonadic.*

Proof. The functor I is Cauchy dense since it is bijective on objects. The result follows by Proposition 9.1. $\square$

It follows that $\Theta: G \Rightarrow H$ induces a functor $\bar{\Theta}: [\mathcal{M}, \mathcal{X}]^G \rightarrow [\mathcal{K}, \mathcal{X}]$ over $[\mathcal{M}, \mathcal{X}]$, where $[\mathcal{M}, \mathcal{X}]^G$ is the category of Eilenberg-Moore G -coalgebras. The composite of $\bar{\Theta}$ with the comparison functor $[\mathcal{J}, \mathcal{X}] \rightarrow [\mathcal{M}, \mathcal{X}]^G$ is isomorphic to

$$\hat{N}: [\mathcal{J}, \mathcal{X}] \longrightarrow [\mathcal{K}, \mathcal{X}] \tag{6.26}$$

as obtained in Theorem 4.1.

Using the finite well-poweredness assumption on the factorization system of $\mathcal{M}$, we shall show that the functor (6.26) is an equivalence when $\mathcal{X}$ is additive. In other words, we shall show that the free additive categories on $\mathcal{J}$ and $\mathcal{K}$ are Morita equivalent (= additively Cauchy equivalent).

Proposition 6.5. *If $\mathcal{X}$ is additive (that is, an Agp-category) then the comonad morphism $\Theta: G \Rightarrow H$ defined at (6.25) is invertible.*

Proof. The ordered set of $\mathcal{M}$ -subobjects of $A \in \mathcal{M}$ is assumed finite and so has a linear refinement. So we can list all the subobjects as $0 = U_0, U_1, \dots, U_n = A$ such that $U_i \leq U_j$ implies $i \leq j$. Then the morphisms (6.25) are represented by an upper-triangular matrix with identity morphisms in the main diagonal. Since $\mathcal{X}$ is additive (including existence of additive inverses), such matrices are invertible. $\square$

Remark 6.6. In case $\mathcal{X}$ is a protomodular 1/Set-category (see Section 8 for references), morphisms $f: A + B \rightarrow A \times B$, with $\text{pr}_1 \circ f \circ \text{in}_1 = 1_A$, $\text{pr}_2 \circ f \circ \text{in}_2 = 1_B$ and $\text{pr}_2 \circ f \circ \text{in}_1 = 0$, are strong epimorphisms. So, for such an $\mathcal{X}$, the comonad morphism $\Theta: G \Rightarrow H$ is a strong epimorphism. We would like this to be the basis of a proof that $\tilde{N}$ (and hence $\tilde{M}$ using Corollary 5.2) is crudely monadic but we do not know how. We will give a proof that $\tilde{M}$ is monadic in Section 8 under the extra Assumption 8.1.

Theorem 6.7. *For any additive category $\mathcal{X}$ with finite direct sums and splitting idempotents, the functor (6.26) is an equivalence of categories*

$$[\mathcal{I}, \mathcal{X}] \simeq [\mathcal{M}, \mathcal{X}] .$$

Proof. By Proposition 6.5, the functor (6.26) becomes the comparison from $[\mathcal{I}, \mathcal{X}]$ into the category of G -coalgebras. So the Theorem is now equivalent to comonadicity of the functor $\text{Lan}_J: [\mathcal{I}, \mathcal{X}] \rightarrow [\mathcal{M}, \mathcal{X}]$. Since the unit of the adjunction $\text{Lan}_J \dashv [J, 1]$ is a pointwise coretraction $FA \rightarrow FA \oplus \bigoplus_{U < A} FU$, and hence a strong monomorphism, the functor Lan_J is conservative. So it remains to prove that Lan_J preserves certain equalizers. In fact, it preserves all finite limits. Since limits in $[\mathcal{M}, \mathcal{X}]$ are formed pointwise, it suffices to see that each

$$\text{ev}_A \circ \text{Lan}_J: [\mathcal{I}, \mathcal{X}] \rightarrow \mathcal{X}$$

preserves finite limits where $\text{ev}_A: [\mathcal{M}, \mathcal{X}] \rightarrow \mathcal{X}$ is evaluation at $A \in \mathcal{M}$. By Lemma 6.1,

$$\text{ev}_A \circ \text{Lan}_J \cong \bigoplus_{U \leq A} \text{ev}_U .$$

Each evaluation ev_U preserves limits so their direct sum does. $\square$

Combining Corollary 5.2 with Theorem 6.7, we obtain:

Theorem 6.8. *In the setting of Section 2, the adjunction (4.10) is an equivalence for any additive category $\mathcal{X}$ in which idempotents split and finite direct sums exist.*

7. EXAMPLES OF THEOREM 6.8

Example 7.1. We begin with a baby version of the Dold-Puppe-Kan Theorem. Let $\text{Pt}\mathcal{X}$ denote the category whose objects are split epimorphisms in $\mathcal{X}$, the morphisms are morphisms of the epimorphisms which commute with the splittings; this is what Bourn [6] calls the category of points in $\mathcal{X}$. Take $\mathbb{P}$ to be the free-living adjunction $\mu^* \dashv \mu$ with identity counit $\mu^* \circ \mu = 1$. So $[\mathcal{P}, \mathcal{X}] \cong \text{Pt}\mathcal{X}$. Let $\mathcal{M}$ consist of all the monomorphisms. Then $\mathcal{R}$ contains only the identities. Theorem 6.8 yields

$$\text{Pt}\mathcal{X} \simeq \mathcal{X} \times \mathcal{X} .$$

This example also shows the necessity of $\mathcal{X}$ having homs enriched in abelian groups (not merely commutative monoids). We need $\mathcal{X}$ to have kernels of split epimorphisms and coproducts already. If we also ask that it have finite products then considering

the split epimorphism $X \times Y \rightarrow Y$ given by the projection, the counit is the canonical map $X + Y \rightarrow X \times Y$, so if this is invertible we have hom enrichment in commutative monoids. Now considering the codiagonal $X + X \rightarrow X$, split by one of the injections, it is not hard to show that 1_X has an additive inverse.

Example 7.2. ([13, 14, 23]) Applying Theorem 6.8 to Example 3.2 yields that $[\Delta_{\perp, \top}, \mathcal{X}]$ is equivalent to the category of chain complexes in $\mathcal{X}$.

Example 7.3. Applying Theorem 6.8 to Example 3.3 yields $[\mathbb{I}, \mathcal{X}] \simeq [\Delta_{\text{inj}}, \mathcal{X}]$, the category of semi-simplicial objects in $\mathcal{X}$.

Here are some examples of the general type described in Example 3.1.

Example 7.4. A *(set) species* in the sense of Joyal [20] is a functor $F : \mathfrak{S} \rightarrow \text{Set}$ where $\mathfrak{S}$ is the category of sets and bijective functions. A *pointed-set species* is a functor $F : \mathfrak{S} \rightarrow 1/\text{Set}$. An *R-module species* is a functor $F : \mathfrak{S} \rightarrow \text{Mod}^R$; the case where R is a field is the basic situation of [21].

Following [10], we write FI for the category of finite sets and injective functions. We then see that $\mathcal{M} = \text{FI}$ and $\mathcal{E} = \mathfrak{S}$, while $\mathcal{P} = \text{FI}_{\#}$, the category of injective partial functions.

Corollary 7.5. [10] *The functor*

$$\widehat{M} : [\mathfrak{S}, \text{Mod}^R] \longrightarrow [\text{FI}_{\#}, \text{Mod}^R]$$

is an equivalence of categories.

Example 7.6. Another example relevant to [22] is the category $\mathcal{A} = \text{FIVect}_{\mathbb{F}_q}$ of finite vector spaces over the field $\mathbb{F}_q$ of cardinality q (a prime power) and injective linear functions. Let $\mathcal{E} = \mathfrak{S}\mathbf{l}_q$ be the category of finite $\mathbb{F}_q$ -vector spaces and linear isomorphisms. Then $\mathcal{P} = \text{FI}_{\#q}$ is the category of finite $\mathbb{F}_q$ -vector spaces and injective partial linear functions.

Corollary 7.7. *The functor*

$$\widehat{M} : [\mathfrak{S}\mathbf{l}_q, \text{Mod}^R] \longrightarrow [\text{FI}_{\#q}, \text{Mod}^R]$$

is an equivalence of categories.

Example 7.8. Here are a few examples of categories $\mathcal{A}$ as in Example 3.1 to which Theorem 6.8 applies with $\mathcal{E}$ the epimorphisms and $\mathcal{M}$ the monomorphisms:

- (a) the category of finite abelian groups and group morphisms;
- (b) the category of finite abelian p -groups and group morphisms;
- (c) the category of finite sets and all functions.

Theorem 6.8 also applies to $\mathcal{M}$ in place of $\mathcal{A}$ in these examples. Then $\mathcal{E}$ is replaced by the groupoid of invertible morphisms in $\mathcal{A}$. In case of example (a), the paper [18] describes the groupoid being represented in $\mathcal{X}$.

Example 7.9. Consider the “algebraic” simplicial category Δ_+ whose objects are all the natural numbers and whose morphisms $\xi : m \rightarrow n$ are order-preserving functions

$$\xi : \{0, 1, \dots, m-1\} \longrightarrow \{0, 1, \dots, n-1\}.$$

Put $\mathcal{A} = \Delta_+^{\text{op}}$. Take $\mathcal{M}$ in $\mathcal{A}$ to consist of the surjections in Δ_+ . Pushouts of surjections along arbitrary morphisms exist in Δ_+ . Then $\mathcal{E} = \Delta_{+\text{inj}}^{\text{op}}$ and $\mathcal{P}$ is the opposite of the category whose morphisms $m \rightarrow n$ are cospans

$$m \xrightarrow{\xi} r \xleftarrow{\sigma} n$$

in Δ_+ with σ surjective. We could also take the “topological” simplicial category Δ (omit the object 0) to obtain a reinterpretation of the preoperads in $\mathcal{X}$ in the sense of [3].

Example 7.10. Here is a rather trivial example involving Δ . Take $\mathcal{A}$ to be the category of non-empty ordinals and morphisms the order-preserving functions which preserve first element. Let $\mathcal{M}$ be the class of morphisms which are inclusions of initial segments. This is part of a factorization system where $\mathcal{E} = \Delta_{\perp \neq \top}$ is the category of ordinals with distinct first and last element and morphisms the order-preserving functions which preserve first and last element. Sometimes $\mathcal{E}$ is called the category of *intervals*; there is a duality isomorphism

$$\mathcal{E} \cong \Delta^{\text{op}}.$$

In this case, not only do we have the equivalence

$$[\mathcal{E}, \mathcal{X}] \simeq [\mathcal{P}, \mathcal{X}]$$

of Theorem 6.8, we actually also have an isomorphism

$$\mathcal{P} \cong \mathcal{E}.$$

Example 7.11. Take $\mathcal{A}$ to be a (partially) ordered set with finite infima and the descending chain condition. Then every morphism is a monomorphism and the strong epimorphisms are equalities. So $\mathcal{E}$ is the discrete category $\text{ob}\mathcal{A}$ on the set of elements of the ordered set. The reader may like to contemplate the case where $\mathcal{A}$ is the set of strictly positive integers ordered by division.

8. WHEN $\mathcal{X}$ IS SEMIABELIAN

Semiabelian categories include the category Grp of (not necessarily abelian) groups and group morphisms. In [7] Dominique Bourn gave a version of the Dold-Puppe-Kan Theorem (Example 7.2) for the case where the codomain category $\mathcal{X}$ was semiabelian. In that case it asserted monadicity of the right adjoint in

$$[\mathcal{D}, \mathcal{X}]_{\text{pt}} \begin{array}{c} \xrightarrow{\widehat{M}} \\ \perp \\ \xleftarrow{\widetilde{M}} \end{array} [\mathcal{P}, \mathcal{X}]. \quad (8.27)$$

In this section, we provide a version of this for $\mathcal{P}$ as in Section 2, not only for Δ^{op} . However, we require an extra assumption related to idempotents.

Assumption 8.1. The maximal proper elements of $\text{Sub}A$ can be listed $m_1, \dots, m_n$ such that the idempotents $c_i = m_i \circ m_i^*$ on A satisfy $c_j \circ c_i \circ c_j = c_j \circ c_i$ for all $i < j$.

Throughout we assume our category $\mathcal{X}$ has zero morphisms (that is, has homs enriched in pointed sets).

We begin by providing a non-additive version of the material at the end of Section 10 on idempotents.

Proposition 8.2. *Suppose the category $\mathcal{X}$ has kernels of idempotents. Let e, f be idempotents on an object A of $\mathcal{X}$. If $e \circ f \circ e = e \circ f$ then the intersection of the kernels of e and f exists.*

Proof. Let $k : K \rightarrow A$ be the kernel of e . Since $e \circ f \circ k = e \circ f \circ e \circ f = 0$, there exists a unique g with $f \circ k = k \circ g$. Then $k \circ g \circ g = f \circ k \circ g = f \circ f \circ k = f \circ k = k \circ g$ and k is a monomorphism. So g is idempotent. Then the kernel $\ell : L \rightarrow K$ of g is easily verified to be the intersection of the kernels of e and f . $\square$

Protomodular categories were defined by Bourn [6]: a category $\mathcal{X}$ (with zero morphisms) is *protomodular* when it is finitely complete and, for each object A , the functor $\ker : \text{Pt}A \rightarrow \mathcal{X}$ is conservative. Here $\text{Pt}A$ is the category whose objects (p, X, s) consist of morphisms $p : X \rightarrow A, s : A \rightarrow X$ with $p \circ s = 1_A$, and whose morphisms $f : (p, X, s) \rightarrow (q, Y, t)$ are morphisms $f : X \rightarrow Y$ such that $q \circ f = p$ and $f \circ s = t$. Also the functor $\ker$ takes (p, X, s) to the kernel of p .

The following property is sometimes [5] taken as the definition of protomodular.

Lemma 8.3. *In a protomodular category, if (p, X, s) is an object of $\text{Pt}A$ and $k : K \rightarrow X$ is the kernel of p then $s : A \rightarrow X, k : K \rightarrow X$ are jointly strongly epimorphic.*

Proof. Suppose $m : Y \rightarrow X$ is a monomorphism and $m \circ u = s, m \circ v = k$ for some u, v . Then $m : (p \circ m, Y, u) \rightarrow (p, X, s)$ is a morphism of $\text{Pt}A$. Using v , we see that m induces an isomorphism between the kernel of $p \circ m$ and K . Since $\ker : \text{Pt}A \rightarrow \mathcal{X}$ is conservative, $m : (p \circ m, Y, u) \rightarrow (p, X, s)$ is invertible. So m is invertible. $\square$

Proposition 8.4. *Let $a_1, \dots, a_n$ be a list of idempotents on an object A of a protomodular category $\mathcal{X}$. Suppose $a_i \circ a_j \circ a_i = a_i \circ a_j$ for $i < j$. Suppose $a_i = m_i \circ m_i^*$ is a splitting of a_i via a subobject $m_i : A_i \rightarrow A$ and retraction $m_i^* : A \rightarrow A_i$. Let $k_i : K_i \rightarrow A$ be the kernel of m_i^* (or equally of a_i). Then the morphisms $m_1, \dots, m_n$ along with the inclusion $\cap_i \ker m_i^* \rightarrow A$ are jointly strongly epimorphic.*

Proof. By Lemma 8.3, for each i , the morphisms $m_i : A_i \rightarrow A$ and $k_i : K_i \rightarrow A$ are jointly strongly epimorphic; we will loosely say “ A_i and K_i cover A ”.

If $i > 1$ then $a_1 \circ a_i \circ k_1 = a_1 \circ a_i \circ a_1 \circ k_1 = 0$, and so $a_i \circ k_1$ lands in K_1 , providing a factorization $a_i \circ k_1 = k_1 \circ a_i^1$. Now a_i^1 is also an idempotent, and, for $1 < i < j$, $k_1 \circ a_i^1 \circ a_j^1 \circ a_i^1 = a_i \circ a_j \circ a_i \circ k_1 = a_i \circ a_j \circ k_1 = k_1 \circ a_i^1 \circ a_j^1$, and so $a_i^1 \circ a_j^1 \circ a_i^1 = a_i^1 \circ a_j^1$. Clearly the splitting A_i^1 of a_i^1 is contained in A_i .

The kernel K_i^1 of a_i^1 is $K_i^1 = K_1 \cap K_i$ since $a_1 \circ x = a_i \circ x = 0$ is equivalent to $x = k_1 \circ y$ and $k_1 \circ a_i^1 \circ y = a_i \circ k_1 \circ y = s_i \circ x = 0$; so in fact $a_i^1 \circ y = 0$.

We know that A may be covered by A_1 and K_1 . By Lemma 8.3 again, we know, for each $i > 1$, that K_1 may be covered by the splitting of A_i^1 and the kernel $K_i^1 = K_1 \cap K_i$ of a_i^1 . Since $A_i^1 \leq A_i$, we see that A may be covered by A_1, A_i , and $K_1 \cap K_i$.

Now continue inductively. $\square$

Theorem 8.5. *Suppose $\mathcal{P}$ is as in Section 2 and $\mathcal{X}$ is protomodular (with zero morphisms) with finite coproducts. Then the components of the adjunction (8.27) are strong epimorphisms.*

Proof. The counit has components

$$\sum_{B \leq_m A} \bigcap_{C \prec_n B} \ker Tn^* \longrightarrow TA.$$

We prove this is a strong epimorphism by induction on the number k of maximal proper subobjects $A_1, \dots, A_k$ of A , with $m_i : A_i \rightarrow A$. The result is clear for $k = 1$. For $k > 1$, consider the following diagram.

$$\begin{array}{ccc} \sum_i \sum_{B \preceq_m A} \bigcap_{C \prec_n B} \ker Tn^* & \xrightarrow{\quad} & \sum_i TA_i \\ \delta \downarrow & & \downarrow \gamma \\ \sum_{B \prec_m A} \bigcap_{C \prec_n B} \ker Tn^* & \xrightarrow{\alpha} FA \xleftarrow{\beta} \bigcap_{C \prec_n A} \ker Tn^* \end{array}$$

The component of the counit is strongly epimorphic if and only if α and β are jointly strongly epimorphic. The top row is a coproduct of components of the counit already known to be strongly epimorphic by induction. So it suffices to show that γ and β are jointly strongly epimorphic. Rewriting the domain of β as

$$\bigcap_{C \prec_n A} \ker Tn^* = \bigcap_{i=1}^k \ker Tm_i^*,$$

we see that Proposition 8.4 applies to yield what we want. $\square$

A category $\mathcal{X}$ is *semiabelian* [19] when it has zero morphisms, is protomodular, is Barr exact, and has finite coproducts. A category is *regular* [1] when it is finitely complete, and has the (strong epimorphism, monomorphism)-factorization system existing and stable under pullbacks. It follows that every strong epimorphism is regular (that is, a coequalizer); see [9] for a proof. A category is *Barr exact* when it is regular and every equivalence relation is a kernel pair.

We mentioned Bourn's category $\text{Pt}\mathcal{X}$ in Example 7.1. We will use the following routine fact.

Lemma 8.6. *If $\mathcal{X}$ is a semiabelian category then the functor $\text{Pt}\mathcal{X} \rightarrow \mathcal{X}$, sending each split epimorphism to its kernel, preserves strong epimorphisms.*

Proof. A strong epimorphism

$$\begin{array}{ccc} X & \xrightarrow{g} & Y \\ s \uparrow & & \uparrow t \\ & p \downarrow & & q \downarrow \\ A & \xrightarrow{f} & B \end{array}$$

in $\text{Pt}\mathcal{X}$ has f and g strong epimorphisms in $\mathcal{X}$. From this it is easily verified that the square involving the downward-pointing arrows is a pushout. Factor the morphism in $\text{Pt}\mathcal{X}$ as

$$\begin{array}{ccccc} X & \xrightarrow{g_1} & Z & \xrightarrow{f_1} & Y \\ s \uparrow & & \uparrow t_1 & & \uparrow t \\ & p \downarrow & & q_1 \downarrow & & q \downarrow \\ A & \xrightarrow{1_A} & A & \xrightarrow{f} & B \end{array}$$

in which the right-hand square involving the downward-pointing arrows is a pullback. The induced morphism $\ker q_1 \rightarrow \ker q$ is invertible. Semiabelian categories are Maltsev [19]. Therefore, as a comparison morphism to the pullback in a pushout square in a Maltsev exact category, g_1 is a strong epimorphism (see Theorem 5.7 of [8]). The

induced morphism $\ker p \rightarrow \ker q_1$ is the pullback of g_1 along $\ker q_1 \rightarrow Z$ and so is a strong epimorphism by regularity. $\square$

Theorem 8.7. *If $\mathcal{P}$ is as in Section 2 and $\mathcal{X}$ is semiabelian then the functor $\widetilde{M}: [\mathcal{P}, \mathcal{X}] \rightarrow [\mathcal{P}, \mathcal{X}]_{\text{pt}}$ of (4.10) is (crudely) monadic.*

Proof. By Theorem 8.5, the right adjoint $\widetilde{M}$ is conservative (since this is logically equivalent to the counit being a strong epimorphism). Since $\mathcal{X}$ is semiabelian, it has coequalizers. Therefore $[\mathcal{P}, \mathcal{X}]$ has coequalizers, so, for crude monadicity [30], it suffices to show that $\widetilde{M}$ preserves coequalizers of reflexive pairs.

Both $[\mathcal{P}, \mathcal{X}]$ and $[\mathcal{P}, \mathcal{X}]_{\text{pt}}$ are semiabelian. A limit-preserving functor between semiabelian categories preserves coequalizers of reflexive pairs provided it preserves strong (= regular) epimorphisms (see Lemma 5.1.12 of [4]).

Let $q: S \rightarrow T$ be a strong epimorphism in $[\mathcal{P}, \mathcal{X}]$. Each $q_A: SA \rightarrow TA$ is a strong epimorphism. We must show each $\widetilde{q}_A: \widetilde{S}A \rightarrow \widetilde{T}A$ is a strong epimorphism.

Recall that $\widetilde{T}A$ is calculated by a sequence of kernels of split epimorphisms. This sequence depends only on the object A and the category $\mathcal{P}$, not on the particular functor T . The desired result follows on repeated application of Lemma 8.6. $\square$

9. APPENDIX A: A GENERAL RESULT

Let $\mathcal{V}$ be a symmetric monoidal closed category which is complete and cocomplete. The tensor product of $A, B \in \mathcal{V}$ is written $A \otimes B$ and the unit for tensor is I , following [24]. For a $\mathcal{V}$ -category $\mathcal{X}$, we also write $A \otimes X$ for the tensor and write $[A, X]$ for the cotensor of $A \in \mathcal{V}$ and $X \in \mathcal{X}$; we have

$$\mathcal{X}(A \otimes X, Y) \cong \mathcal{V}(A, \mathcal{X}(X, Y)) \cong \mathcal{X}(X, [A, Y]) .$$

A $\mathcal{V}$ -functor $H: \mathcal{C} \rightarrow \mathcal{A}$ is *Cauchy dense* (as a morphism $H_* = \mathcal{A}(1_{\mathcal{A}}, H): \mathcal{C} \rightarrow \mathcal{A}$ in the bicategory $\mathcal{V}\text{-Mod}$) when the morphism

$$\int^{\mathcal{C}} \mathcal{A}(HC, A_1) \otimes \mathcal{A}(A, HC) \rightarrow \mathcal{A}(A, A_1) \quad (9.28)$$

induced by composition in $\mathcal{A}$ is a strong epimorphism in $\mathcal{V}$ for all $A, A_1 \in \mathcal{A}$. When $\mathcal{V}$ is the category Agp of abelian groups this means that each $A \in \mathcal{A}$ is a retract of a finite direct sum of objects HC in the image of H . The following result is standard but we provide a proof.

Proposition 9.1. *Suppose $\mathcal{X}$ is a cocomplete $\mathcal{V}$ -category, $\mathcal{C}$ is a small $\mathcal{V}$ -category, and $H: \mathcal{C} \rightarrow \mathcal{A}$ is a Cauchy dense $\mathcal{V}$ -functor. Then the $\mathcal{V}$ -functor $[H, 1]: [\mathcal{A}, \mathcal{X}] \rightarrow [\mathcal{C}, \mathcal{X}]$ is both monadic and comonadic, and so preserves and reflects limits, colimits, and strong epimorphisms, as well as being conservative.*

Proof. The left adjoint to $[H, 1]$ is left Kan extension Lan_H along H defined by the coend formula

$$\text{Lan}_H(G) = \int^{\mathcal{C}} \mathcal{A}(HC, -) \otimes GC .$$

Using the coend form of the Yoneda lemma (see [12]), we see that the component at $F \in [\mathcal{A}, \mathcal{X}]$ of the counit for the adjunction $\text{Lan}_H \dashv [H, 1]$ is isomorphic to the

morphism

$$\int^{C,A} (\mathcal{A}(HC, -) \otimes \mathcal{A}(A, HC)) \otimes FA \longrightarrow \int^A \mathcal{A}(A, -) \otimes FA$$

induced by (9.28). Since H is Cauchy dense, these components are all strong epimorphisms. It follows that the right adjoint $[H, 1]$ is conservative. It is clear that $[H, 1]$ preserves colimits. Since colimits exist in $[\mathcal{A}, \mathcal{X}]$, it follows that $[H, 1]$ also reflects colimits. By an easy case of the Beck monadicity theorem [29], since $[H, 1]$ has a left (respectively, right) adjoint, is conservative and preserves coequalizers (respectively, equalizers), it is monadic (respectively, comonadic). $\square$

Suppose $M: \mathcal{B} \rightarrow \mathcal{A}$ is a $\mathcal{V}$ -module viewed as a $\mathcal{V}$ -functor $M: \mathcal{A}^{\text{op}} \otimes \mathcal{B} \rightarrow \mathcal{V}$. We call M the *kernel* of the adjunction

$$[\mathcal{A}, \mathcal{X}] \xrightleftharpoons[\widetilde{M}]{\widehat{M}} [\mathcal{B}, \mathcal{X}] , \quad (9.29)$$

where

$$\widetilde{M}(T)A = \int_B [M(A, B), TB] \quad (9.30)$$

and

$$\widehat{M}(F)B = \int^A M(A, B) \otimes FA . \quad (9.31)$$

The counit and unit of (9.29) will be denoted by

$$\varepsilon_M: \widehat{M}\widetilde{M} \Longrightarrow 1_{[\mathcal{B}, \mathcal{X}]} \quad \text{and} \quad \eta_M: 1_{[\mathcal{A}, \mathcal{X}]} \Longrightarrow \widetilde{M}\widehat{M} .$$

To say $M: \mathcal{B} \rightarrow \mathcal{A}$ is an equivalence as a $\mathcal{V}$ -module is the same as saying $\widetilde{M}: [\mathcal{B}, \mathcal{V}] \rightarrow [\mathcal{A}, \mathcal{V}]$ is an equivalence of $\mathcal{V}$ -categories. We also say $\mathcal{A}$ and $\mathcal{B}$ are *Cauchy* (or *Morita*) equivalent when such an equivalence exists. It follows that $\mathcal{A}^{\text{op}}$ and $\mathcal{B}^{\text{op}}$ are also Cauchy equivalent, and that the adjunction (9.29) is an equivalence for any Cauchy complete $\mathcal{X}$. See [38] for more detail.

Suppose $M: \mathcal{B} \rightarrow \mathcal{A}$ and $N: \mathcal{D} \rightarrow \mathcal{C}$ are $\mathcal{V}$ -modules, and $H: \mathcal{C} \rightarrow \mathcal{A}$ and $K: \mathcal{D} \rightarrow \mathcal{B}$ are $\mathcal{V}$ -functors. Each $\mathcal{V}$ -natural family

$$\theta_{C,D}: N(C, D) \rightarrow M(HC, KD) \quad (9.32)$$

induces $\mathcal{V}$ -natural families $\theta_{A,D}^r$:

$$\int^C N(C, D) \otimes \mathcal{A}(A, HC) \xrightarrow{\int^C \theta \otimes 1} \int^C M(HC, KD) \otimes \mathcal{A}(A, HC) \xrightarrow{\text{act}^r} M(A, KD)$$

and $\theta_{C,B}^\ell$:

$$\int^D \mathcal{B}(KD, B) \otimes N(C, D) \xrightarrow{\int^D 1 \otimes \theta} \int^D \mathcal{B}(KD, B) \otimes M(C, D) \xrightarrow{\text{act}^\ell} M(HC, B) .$$

We obtain three $\mathcal{V}$ -natural transformations (9.33), all mates under appropriate adjunctions.

$$\begin{array}{ccc}
[\mathcal{A}, \mathcal{X}] & \xrightarrow{[H,1]} [\mathcal{C}, \mathcal{X}] & \xrightarrow{\text{Lan}_H} [\mathcal{A}, \mathcal{X}] \\
\widehat{M} \downarrow & \xleftarrow{\theta^r} \widehat{N} \downarrow & \xrightarrow{\theta^\ell} \downarrow \widehat{M} \\
[\mathcal{B}, \mathcal{X}] & \xrightarrow{[K,1]} [\mathcal{D}, \mathcal{X}] & \xrightarrow{\text{Lan}_K} [\mathcal{B}, \mathcal{X}]
\end{array}
\quad
\begin{array}{ccc}
[\mathcal{B}, \mathcal{X}] & \xrightarrow{[K,1]} [\mathcal{D}, \mathcal{X}] \\
\widetilde{M} \downarrow & \xrightarrow{\tilde{\theta}} \widetilde{N} \downarrow \\
[\mathcal{A}, \mathcal{X}] & \xrightarrow{[H,1]} [\mathcal{C}, \mathcal{X}]
\end{array}
\quad (9.33)$$

Notice that $\tilde{\theta}$ is invertible if and only if θ^ℓ is invertible. Also, we have the following two commutative squares.

$$\begin{array}{ccc}
\widehat{N}[H, 1]\widetilde{M} & \xrightarrow{\theta^r \widetilde{M}} [K, 1]\widehat{M}\widetilde{M} & [H, 1] \xrightarrow{\eta_N[H, 1]} \widetilde{N}\widehat{N}[H, 1] \\
\downarrow \widehat{N}\tilde{\theta} & \downarrow [K, 1]\varepsilon_M & \downarrow [H, 1]\eta_M \\
\widehat{N}\widetilde{N}[K, 1] & \xrightarrow{\varepsilon_N[K, 1]} [K, 1] & [H, 1]\widehat{M}\widetilde{M} \xrightarrow{\tilde{\theta}\widehat{M}} \widetilde{N}[K, 1]\widehat{M}
\end{array}
\quad (9.34)$$

Proposition 9.2. *Suppose $\mathcal{X}$ is a cocomplete $\mathcal{V}$ -category, the $\mathcal{V}$ -functors $H: \mathcal{C} \rightarrow \mathcal{A}$ and $K: \mathcal{D} \rightarrow \mathcal{B}$ are Cauchy dense with $\mathcal{C}$ and $\mathcal{D}$ small, and $\theta: N \Rightarrow M(H, K)$ is as in (9.32).*

- (a) *If $\tilde{\theta}$ is a strong epimorphism and $\widetilde{N}$ is conservative then $\widetilde{M}$ is conservative.*
- (b) *If θ^r is a strong monomorphism and $\widehat{N}$ is conservative then $\widehat{M}$ is conservative.*
- (c) *If $\tilde{\theta}$ is invertible and $\widetilde{N}$ preserves reflexive coequalizers then $\widetilde{M}$ preserves reflexive coequalizers.*
- (d) *If θ^r is invertible and $\widehat{N}$ preserves reflexive coequalizers then $\widehat{M}$ preserves reflexive coequalizers.*
- (e) *If $\tilde{\theta}$ and θ^r are invertible and $\widetilde{N}$ is fully faithful then $\widetilde{M}$ is fully faithful.*
- (f) *If $\tilde{\theta}$ and θ^r are invertible and $\widehat{N}$ is fully faithful then $\widehat{M}$ is fully faithful.*

Proof. For (a), contemplate the left square of (9.34). If $\tilde{\theta}$ is a strong epimorphism, so too is $\widehat{N}\tilde{\theta}$. If $\widetilde{N}$ is conservative then $\varepsilon_N[K, 1]$ is a strong epimorphism. It follows that $[K, 1]\varepsilon_M$ is a strong epimorphism. By Proposition 9.1, ε_M is a strong epimorphism. So $\widetilde{M}$ is conservative.

For (b), contemplate the right square of (9.34) and use the dual argument.

For (c), look at the rightmost square of (9.34) with its invertible 2-cell. Take any kind of colimit in $[\mathcal{B}, \mathcal{X}]$. It is preserved by $[K, 1]$ and so, if that kind of colimit is preserved by $\widetilde{N}$, then $[H, 1]\widetilde{M}$ preserves the colimit. By Proposition 9.1, $[H, 1]$ reflects the colimit and we have the result.

For (d), look at the leftmost square of (9.34) and apply the dual argument.

For (e), contemplate the left square of (9.34) and deduce from our hypotheses that $[K, 1]\varepsilon_M$ is invertible. By Proposition 9.1, ε_M is invertible, so $\widetilde{M}$ is fully faithful.

For (f), contemplate the right square of (9.34) and use the dual argument. $\square$

We shall say a $\mathcal{V}$ -functor is *crudely monadic* when it has a left adjoint, is conservative, has reflexive coequalizers existing in the domain, and preserves reflexive coequalizers.

Corollary 9.3. *Suppose H and K are Cauchy dense and both θ^ℓ and θ^r are invertible.*

- (a) If $\tilde{N}: [\mathcal{D}, \mathcal{V}] \rightarrow [\mathcal{C}, \mathcal{V}]$ is crudely monadic then $\tilde{M}: [\mathcal{B}, \mathcal{V}] \rightarrow [\mathcal{A}, \mathcal{V}]$ is too.
 (b) If $\tilde{N}: [\mathcal{D}, \mathcal{V}] \rightarrow [\mathcal{C}, \mathcal{V}]$ is an equivalence then $\tilde{M}: [\mathcal{B}, \mathcal{V}] \rightarrow [\mathcal{A}, \mathcal{V}]$ is too.

10. APPENDIX B: REMARKS ON IDEMPOTENTS

Define a relation on any monoid M by $a \sqsubseteq b$ when $ba = a$. This relation is transitive; indeed, we have a stronger property.

Proposition 10.1. *If $ub \sqsubseteq a$ and $vc \sqsubseteq b$ then $uvc \sqsubseteq a$.*

If $u_i a_i \sqsubseteq a_{i-1}$ for $1 \sqsubseteq i \sqsubseteq n$ then $u_1 \dots u_n a_n \sqsubseteq a_0$.

Proof. For the first sentence, the assumptions are $aub = ub$ and $bvc = vc$. So $auvc = aubvc = ubvc = uvc$; that is, $uvc \sqsubseteq a$. The second sentence follows by induction. $\square$

The relation is only reflexive for idempotents: clearly $a \sqsubseteq a$ is equivalent to $aa = a$.

The unit 1 of the monoid is a largest element in the sense that $a \sqsubseteq 1$ for all $a \in M$. That is, 1 is the empty meet. However, not all meets need exist. A *meet* for $a, b \in M$ is an element $a \wedge b$ with $a \wedge b \sqsubseteq a$ and $a \wedge b \sqsubseteq b$, and, if $x \sqsubseteq a$ and $x \sqsubseteq b$ then $x \sqsubseteq a \wedge b$. In particular, $a \wedge b$ must be an idempotent. Meets of lists of n elements are defined in the obvious way and, for $n \geq 2$, can be constructed from iterated binary meets when they exist.

Proposition 10.2. *If $ab \sqsubseteq b$ and $a \sqsubseteq a$ then $a \wedge b = ab$.*

If $a_1, \dots, a_n$ are idempotents such that $a_i a_j \sqsubseteq a_j$ for $i \sqsubseteq j$ then $a_1 \wedge \dots \wedge a_n = a_1 \dots a_n$.

Proof. For the first sentence, we are told that $ab \sqsubseteq b$, while $a \sqsubseteq a$ implies $aa = a$, and so $aaab = ab$, yielding $ab \sqsubseteq a$. For the second sentence the result is clear for $n = 1$ since we suppose a_1 idempotent. Assume the result for $n - 1$; so $a_1 \wedge \dots \wedge a_{n-1} = a_1 \dots a_{n-1}$. Apply the second sentence of Proposition 10.1 to the inequalities $a_i a_n \sqsubseteq a_n$ to deduce $a_1 \dots a_{n-1} a_n \sqsubseteq a_n$. So, by the first sentence, $a_1 \dots a_n = a_1 \dots a_{n-1} \wedge a_n = a_1 \wedge \dots \wedge a_{n-1} \wedge a_n$, as required. $\square$

Notice that, if $ab = ba$ and b is idempotent, then $bab = abb = ab$, so $ab \sqsubseteq b$. So the proposition applies to commuting idempotents.

Now suppose we have a ring R . We can apply our results to the multiplicative monoid of R . We say idempotents e and f in R are *orthogonal* when $ef = fe = 0$. A list $e_0, e_1, \dots, e_n$ of idempotents is *orthogonal* when each pair in the list is orthogonal. The list is *complete* when $e_0 + e_1 + \dots + e_n = 1$. An easy induction shows that a complete list of idempotents is orthogonal if and only if $e_i e_j = 0$ for $i \leq j$.

For each $a \in R$, put $\bar{a} = 1 - a$. Clearly if a is idempotent, so is $\bar{a}$.

Let R° denote the ring obtained from R by reversing multiplication.

Proposition 10.3. (a) $a \sqsubseteq b$ in R if and only if $\bar{b} \sqsubseteq \bar{a}$ in R° .

(b) For b an idempotent, $ab \sqsubseteq b$ in R if and only if $\bar{a}\bar{b} \sqsubseteq \bar{b}$ in R° .

(c) If a and b are idempotents and $ab \sqsubseteq b$ in R then $e_0 = ab, e_1 = \bar{a}b, e_2 = \bar{b}$ is a complete list of orthogonal idempotents.

Proof. (a) $\bar{b} \sqsubseteq \bar{a}$ in R° means $(1 - b)(1 - a) = 1 - b$ in R ; that is, $ba = a$ which means $a \sqsubseteq b$ in R .

- (b) $\bar{a}\bar{b} \sqsubseteq \bar{b}$ in R° means $\bar{b}\bar{a}\bar{b} = \bar{b}\bar{a}$ in R . That is, $(1-b)(1-a)(1-b) = (1-b)(1-a)$. That is, $1 - a - b + ab - b + ba + bb - bab = 1 - b - a + ba$. That is, $bab = ab$, which is $ab \sqsubseteq b$ in R .
- (c) We already know e_0 and e_2 are idempotent. They are also orthogonal: $e_0e_2 = ab(1-b) = ab - ab = 0$ and $e_2e_0 = (1-b)ab = ab - bab = 0$. Therefore $e_0 + e_2 = ab + 1 - b = 1 - (1-a)b = \bar{e}_1$ is idempotent. So e_1 is idempotent and $e_0 + e_1 + e_2 = 1$. The calculations $e_0e_1 = ab(1-a)b = ab - abab = 0$ and $e_1e_2 = (1-a)b(1-b) = b - ab - b + abb = 0$ complete the proof. $\square$

We can extend part (c) inductively to obtain:

Proposition 10.4. *Suppose $a_1, \dots, a_n$ are idempotents such that $a_i a_j \sqsubseteq a_j$ for $i \leq j$. Then $e_i = \bar{a}_i a_{i+1} a_{i+2} \dots a_n$ for $0 \leq i \leq n$ (in particular, $e_0 = a_1 a_2 \dots a_n$ and $e_n = \bar{a}_n$) defines a complete list of orthogonal idempotents.*

Suppose $\mathcal{X}$ is an additive category in which idempotents split. Our results apply to the endomorphism monoid $\mathcal{X}(A, A)$ of each object $A \in \mathcal{X}$. If a is an idempotent on A , we have a splitting:

$$\begin{array}{ccc} A & \xrightarrow{a} & A \\ r_a \downarrow & \nearrow i_a & \downarrow r_a \\ aA & \xrightarrow{1_{aA}} & aA \end{array}$$

Yet, we also have a splitting for $\bar{a} = 1 - a$ which incidentally provides a kernel $\bar{a}A$ for a and so a direct sum decomposition of A :

$$A \cong \bar{a}A \oplus aA.$$

More generally, for any complete list $e_0, e_1, \dots, e_n$ of orthogonal idempotents in $\mathcal{X}(A, A)$, we obtain a direct sum decomposition

$$A \cong e_0A \oplus e_1A \oplus \dots \oplus e_nA.$$

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